

Sprint 3

Sprint Goal: Complete the backend workflow engine by implementing lifecycle management, expiry rules, lifecycle reset, audit functionality and automated container simulation while refining the applications business rules.

Sprint Backlog

ID	User Story	Priority	Story Points	Status
PB-10	Production Lifecycle	Must Have	8	Done
PB-11	Lifecycle Expiry reset	Must Have	5	Done
PB-12	Audit trail	Should Have	3	Done
PB-13	Automated container simulation	Should Have	8	Done

Development Tasks

- Refine container workflow logic
- Resolve approval workflow issues
- Implement production lifecycle management
- Implement lifecycle expiry based on usage and elapsed time
- Implement automatic lifecycle reset following supervisor approval
- Record complete workflow audit information
- Develop automated container lifecycle simulation
- Simulate production batches and container movements
- Validate lifecycle behavior through simulation

GitHub Evidence

- 2 July 2026- Complete first container lifecycle with QA approval
- 2 July 2026-Complete initial production approval workflow
- 3 July 2026- Fixed Container workflow, approval logic, lifecycle updates, and cleaned service files
- 4 July 2026-Complete production lifecycle with expiry, approvals and expiry reset
- 5 July 2026-Backend Workflow complete: lifecycle management, A approval, supervisor approval workflow, lifecycle expiry/reset and audit trail
- 5 July 2026- Complete backend workflow engine
- 6 July 2026-Complete automated container lifecycle simulation
- 7 July 2026- Complete backend workflow engine and simulation

Definition of Done

- Production lifecycle implemented
- Automatic expiry rules operational
- Automatic lifecycle reset implemented
- Audit trail operational
- Automated simulation complete
- Business rules validated
- Features tested using Thunder Client and database verification
- Source code committed to GitHub
- Sprint deliverables integrated successfully

Sprint review

Sprint 3 completed the core backend implementation of the Container Management System. The production lifecycle was extended to support lifecycle expiry-based usage limit and elapsed time. The supervisor's approval enables the lifecycle reset and continued use of containers. An automated simulation engine was also developed to generate realistic operational data and validate the workflow under multiple lifecycle scenarios. By the end of the sprint, the backend workflow engine had been fully implemented and successfully validated through simulation.

What Went Well

- Backend workflow engine completed successfully
- Lifecycle expiry and reset rules implemented
- Automated simulation validated business rules
- Workflow service became more stable after refactoring
- Audit functionality improves traceability

Challenges

- Refining complex lifecycle business rules
- Maintaining consistency between expiry, approval and reset logic
- Debugging workflow during simulation
- Keeping workflow engine maintained as new functions were added

Focus for Sprint 4

- Comprehensive backend testing
- Integration testing
- React frontend development
- Dashboard Implementation
- Documentation and UML diagrams